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SAFETY DATA SHEET

1. CHEMICAL PRODUCT AND COMPANY IDENTIFICATION

Product name	N-n-Butoxymethylacrylamide(Chemical name abbreviation:NBMA)	
Product code	mucAE_ NBMA	
Reference number	mucE3276-13	
Supplier name	Shinryo Corporation	
Address	3-9-22, Kurosaki, Yahatanishi-ku, Kitakyushu-City, Fukuoka, Japan	
Department	Marketing Division	
Phone	+81-93-643-2778	
Emergency telephone	Week day / Day time; Hachinohe Plant Coordination Section +81-178-44-1112	
	Holiday / Night time; Hachinohe Plant Production Section +81-178-44-1114	
FAX	+81-93-643-2061	
Recommended use and restricted use	Raw material for paint and adhesive agent	

2. HAZARD IDENTIFICATION

GHS Classification

Health hazards	Acute toxicity (oral)	Category 4
	Acute toxicity (dermal)	Category 3
	Serious eye damage/eye irritation	Category 2B
	Germ cell mutagenicity	Category 1B
	Carcinogenicity	Category 1B
	Toxic to reproduction	Category 1B
	Specific target organ toxicity (single exposure)	Category 1
		(systemic toxicity)

The hazards not listed above are classified either as Not applicable or Classification not possible.

GHS label elements

Pictograms



Signal word

Hazard statements

Danger

H302	Harmful if swallowed
H311	Toxic in contact with skin
H340	May cause genetic defects
H350	May cause cancer
H360	May damage fertility or the unborn child
H370	Causes systemic toxicity

Precautionary statements

Safe handling

Do not breathe dust/fume/gas/mist/vapours/spray.(P260)

Emergency measure	Wash the hands thoroughly after handing.(P264)
	Obtain special instructions before use.(P201)
	Do not handle until all safety precautions have been read and understood.(P202)
	Wash the hands thoroughly after handling. (P264)
	Do not eat,drink or smoke when using this product.(P270)
	Wear protective gloves/protective clothing.(P280)
	If in eyes, rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing. (P305+P351+P338)
	Specific treatment is necessary (see Section 4, “First aid measures” for the details). (P321)
	If eye irritation persists, get medical advice/attention. (P337+P313)
	Take off immediately all contaminated clothing and wash it before reuse.(P361+P364)
Storage	IF ON SKIN:Wash with plenty of soap and water.(P302+P352)
	IF exposed or concerned:Get medical advise/attention.(P308+P313)
Disposal	Call a POISON CENTER or doctor/physician if you feel unwell. (P312)
	Specific treatment. (see Section 4, “First aid measures” for the details) (P322)
	Rinse mouth.(P330)
	Store locked up.(P405)
	Dispose of contents/container by contracting an agency for industrial waste disposal licensed by the prefectural governor. (P501)

3.COMPOSITION AND INFORMATION ON INGREDIENTS

Chemical substance/mixture

Mixture

Chemical name or generic name

N-n-Butoxymethylacrylamide

Chemical name or generic name	Concentration or concentration range	Chemical formula	CAS RN	EINECS number	TSCA
N-n-Butoxymethylacrylamide	95%～98%	C ₈ H ₁₅ NO ₂	1852-16-0	217-442-7	1852-16-0
Acrylamide	0.7%	C ₃ H ₅ NO	79-06-1	201-173-7	79-06-1
N-Methylolacrylamide	0.1%～0.5%	C ₄ H ₇ NO ₂	924-42-5	213-103-2	924-42-5
n-Butanol	0.2%～0.4%	C ₄ H ₁₀ O	71-36-3	200-751-6	71-36-3

Impurities and stabilizing additives which No information
contribute to the classification of the substance

4.FIRST AID MEASURES

If inhaled	Remove victim to fresh air and keep at rest in a position comfortable for breathing. Call a doctor/physician if you feel unwell. If breathing is stopped, give artificial respiration.
If on skin	Remove or take off immediately all contaminated clothing. Rinse skin with running water/shower. Wash skin with plenty water and soap. Get medical attention immediately if pain or inflammation is noted.
If in eyes	Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do.

If swallowed	Continue rinsing. Get medical advice/attention. Widely open the eyelid when rinsing so that eyeball and whole eye lid can be washed thoroughly. Immediately call a doctor/physician. If the victim is conscious, wash inside mouth thoroughly with water. Do not give anything by mouth to the victim when not conscious.
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5.FIRE-FIGHTING MEASURES

Suitable extinguishing media	Dry chemical (powder), carbon dioxide, water spray, sand, general foam fire extinguisher.
Extinguishing media that must not be used	Straight spray water may enlarge fire by spreading the liquid.
Specific hazards arising from the fire	Heating may induce explosion of containers. Fire may produce irritant/corrosive/toxic gases. Explosive polymerization may be caused if involved in fire.
Specific fire-fighting method	Remove container, packaging and such to a safe place immediately if possible in case of fire in the vicinity. Cool unremovable container, packaging, and surrounding facilities by sprinkling water. Fully cool containers with plenty of water even after extinction.
Protection for fire-fighter	Extinguish from the windward. Wear respiratory protection if required.

6.ACCIDENTAL RELEASE MEASURES

Personal precautions, protective equipment and emergency procedures	Ventilate before entering tightly closed places. Do not allow non-authorized personnel to access around the leakage area: mark the area using rope, etc. Appropriate protective equipment (see “8. Exposure controls/personal protection) must be worn to prevent contact with eye or skin and inhalation. Work from the upwind position and evacuate people in the leeward to the safe place.
Environmental precautions	Do not allow spillage out into the environment causing pollution e.g., rivers, waterways. If diluting with large amount of water, do not allow polluted discharge water out into the environment without appropriate treatment. Immediately report to the relevant authorities if released to rivers, waterways.
Methods and materials for containment and cleaning up	Block the source of the leak if safe to do so. Prevent leakage/release of the product. In case of small spillage absorb with sand or non-flammable materials or collect into empty container that can be tightly closed and sealed. In case of large spillage, contain by making on embankment of earth to prevent releasing, lead the spillage to a safe place and collect.
Prevention of secondary disaster	Remove all potential fire sources immediately, and prepare fire extinguisher for ignition.

7.HANDLING AND STORAGE

Handling

Engineering measures	Take facility measures as described in “8. Exposure controls/personal protection” and wear protective equipment. Work under the environment maintained in accordance with the regulations under Fire Service Act and Ordinance on Prevention of Organic Solvent Poisoning. Keep away from fire. Do not heat the product excessively, do not give friction or impact. Avoid contact with strong oxidizer. Ensure grounding the facilities and equipment for electrostatic countermeasure. Use explosion-proof electrical equipment. Use electroconductive work clothes and shoes. Do not use high-temperature material, sparks or fire near the product. Use anti-spark tools. Set local exhaust or general ventilation as described in “8. Exposure controls/personal protection”.
Precautions for safe handling	Obtain special instructions before use. Do not handle until all safety precautions have been read and understood.
Contact avoidance	Wash the hands thoroughly after handling.
Hygienic measures	See “10. Stability and reactivity”. Wash the hands thoroughly after handling. Contaminated work clothing should not be allowed out of the workplace.
Storage	
Safe storage conditions	In the store room, install the daylighting, lighting, and ventilating equipment needed for storing or handling hazardous substances. Avoid contact with metals, peroxides and such materials that promote polymerization as the product is polymerizable. Storage conditions should be in compliance with the designation under Fire Service Act. Keep in indoor storehouse for hazardous goods.
Safe container material	Packaging or container is not specified but stainless container that can be tightly closed and sealed is recommended. Stainless container that can be tightly closed and sealed is recommended. Resin container or hybrid can (inner coated stainless container) that can be tightly closed and sealed is recommended. Follow the regulations of Fire Service Act, Act on Port Regulations, Civil Aeronautics Act and UN transport regulations.

8.EXPOSURE CONTROLS / PERSONAL PROTECTION

	Allowable exposure limit (ACGIH)
n-Butanol	TWA 20 ppm, STEL -
Acrylamide	TWA 0.03 mg/m3 (IFV), STEL – (Skin)

Facility measures	Enclosed facility, equipment or local exhaust should be used for handling. Establish eye-, hand-washing facilities and safety shower in the work areas for this product, clearly indicate the location. Use explosion-proof electrical/ventilating/lighting equipment.
Protective equipments	
Respiratory protection	Use personal respiratory protection as required. Use organic gas cartridge on gas mask.
Hand protection	Wear protective gloves. Protective gloves.
Eye protection	Use personal eye protection as required. Protective glasses (normal type, normal type with side shields, goggle type)
Skin and body protection	Use personal protective clothing, face protection as required.

9. PHYSICAL AND CHEMICAL PROPERTIES

Appearance

Form	Liquid
Color	Pale yellow
Odor	Characteristic odor
Odor threshold	No data available.
pH	No data available.
Melting point/freezing point	-9 to -6 °C
Boiling point, initial boiling point and boiling range	110 to 112 °C/4Pa
Flash point	121 °C (closed cup)
Evaporation rate	No data available.
Combustibility (solid, gas)	No data available.
Flammability or explosive limits	
Lower limit	No data available.
Upper limit	No data available.
Vapor pressure	No data available.
Vapor density	No data available.
Specific gravity (relative density)	0.977(25 °C)
Solubility	Insoluble in water.
n-Octanol/water partition coefficient	No data available.
Auto-ignition point	No data available.
Decomposition point	No data available.
Viscosity (viscosity coefficient)	No data available.

As n-Butanol

Melting point/freezing point	-89 °C
Boiling point, initial boiling point and boiling range	117.5 °C
Flash point	35 to 37.8 °C (Tag closed cup)
Flammability or explosive limits	
Lower limit	1.4 vol%
Upper limit	11.2 vol%
Vapor pressure	About 0.6 kPa (20°C)

Vapor density	(Air = 1) 2.6
n-Octanol/water partition coefficient	0.88
As acrylamide	
Melting point/freezing point	84 to 86 °C
Boiling point, initial boiling point and boiling range	125 °C/25mmHg;87 °C 2mmHg;
Flash point	138 °C
Vapor pressure	120 Pa (25 °C)

10.STABILITY AND REACTIVITY

Reactivity	No information available.
Chemical stability	Considered to be stable under storage and handling conditions in accordance with the laws and regulations.
Possibility of hazardous reactions	Polymerization may be caused at high temperature or under direct sunlight.
Conditions to avoid	Avoid high temperature.
Incompatible materials	No information available.
Hazardous decomposition products	No information available.

11.TOXICOLOGICAL INFORMATION

As N-n-Butoxymethyl acrylamide

Acute toxicity: oral	LD50, rat = 999.6 mg/kg, 630 mg/kg (US EPA/HPV Challenge program (2001)). Based on this the substance is classified as Category 4.
Acute toxicity: dermal	LD50, rat = 866 mg/kg (US EPA/HPV Challenge program (2001)). Based on this the substance is classified as Category 3.
Acute toxicity: inhalation (gas)	The substance is liquid under GHS criteria.
Acute toxicity: inhalation (vapour)	No data available.
Acute toxicity: inhalation (dust)	No data available.
Acute toxicity: inhalation (mist)	No data available.
Skin corrosion/irritation	Slight to clear erythema was noted at 24 hours after occlusive application of 0.1 mL of the substance to rabbit skin in a Draize test. The symptom disappeared by 72 hours, no oedema was noted in intact skin and primary irritation index was 1.8 (US EPA/HPV Challenge program (2001)). Based on this the substance is not classified according to JIS classification criteria (corresponds to Category 3 of UN GHS classification).
Serious eye damage/eye irritation	Corneal damage was noted in 3 out of 6 animals after instillation of 0.5 mL of the substance to rabbit eye in a Draize test, and the result was "mild irritant" (US EPA/HPV Challenge program (2001)). Based on this the substance is classified as Category 2B.
Respiratory sensitization or skin sensitization	Respiratory sensitization: No data available. Skin sensitization: No data available.
Germ cell mutagenicity	Classification is not possible due to no data of in vivo study. As in vitro study negative result of Ames test and positive result of chromosome aberration study using ovary cells of Chinese hamster are reported (US EPA/HPV Challenge program (2001)).
Carcinogenicity	No data available.
Reproductive toxicity	No data available.
Specific target organ toxicity (single exposure)	As a result of dermal administration study in the rabbit, tremor, congestion in lung, pale kidney were noted in dead animals and LD50 was 866.3 mg/kg (US EPA/HPV Challenge program (2001)), which

Specific target organ toxicity (repeated exposure)	corresponds to Category 1 level. In an acute oral toxicity study in the rat, most animals showed gasping and inanimation, and hemorrhage in lung, gastrointestinal were observed in dead animals and congestion in adrenal gland and kidney in survived animals in necropsy, LD50 was 999.6 mg/kg (US EPA/HPV Challenge program (2001)), which corresponds to Category 2 level. Based on these it is difficult to specify the target organ, but the substance is classified as Category 1 (systemic toxicity) as dermal LD 50 in rabbit corresponds to Category 1 level. Classification is not possible due to lack of data. No remarkable effect is reported other than mild ataxia in high dose group (296 - 306 mg/kg/day: 90-day conversion, 138 - 143 mg/kg/day) in 42-day dietary administration study in the rat, but pathological findings and such details of the test results are unknown.
Aspiration hazard	No data available.
As n-Butanol	
Acute toxicity	Inhalation, LC50, rat, 4H = 8000 ppm Oral, LD50, rat = 2500 mg/kg Oral, LD50, rabbit = 3484 mg/kg
Acute toxicity: oral	LD50, rat = 790 mg/kg (vol.3, Risk Assessment, Ministry of the Environment (2004), SIDS (2004), EHC 65 (1987)), 4360 mg/kg (SIDS (2004), ACGIH (2002), PATTY (4th, 1994)), 2290 mg/kg (SIDS (2004)), 2510 mg/kg (SIDS (2004), ACGIH (2002), PATTY (4th, 1994)), 2100 mg/kg (EHC 65 (1987), DFGOT vol.19 (2003)), 700 mg/kg (EHC 65 (1987), DFGOT vol.19 (2003)). Based on these calculation was used for classification and the result was 1227 mg/kg. Therefore the substance is classified as Category 4.
Acute toxicity: dermal	LD50, rabbit = 3402 mg/kg (SIDS (2004), DFGOT vol.19 (2003)), 5300 mg/kg (SIDS (2004), EHC 65 (1987), ACGIH (2002), DFGOT vol.19 (2003), PATTY (4th, 1994)), 4200 mg/kg (EHC 65 (1987), PATTY (4th, 1994)). Based on these calculation was used for classification and the result was 3636 mg/kg. Therefore the substance is classified as Category 5.
Acute toxicity: inhalation (gas)	The substance is liquid under GHS criteria.
Acute toxicity: inhalation (vapour)	Saturated vapor pressure concentration is 5726 ppm based on vapor pressure = 0.58kPa (20°C), rat LC50 (4H) > 24 mg/L (NITE) = 7917 ppm, 7917 ppm > 5726 ppm. Based on this the classification is examined for 'inhalation (mist)' and classification for 'inhalation (vapour)' is not possible due to no data.
Acute toxicity: inhalation (dust)	No data available.
Acute toxicity: inhalation (mist)	No data available.
Skin corrosion/irritation	Moderate irritation in 24-hour patch test using the rabbit is reported in SIDS (2004), EHC 65 (1987) and DFGOT vol.19 (2003), dermatitis in occupational exposure is reported in SIDS (2004), ACGIH (2002) and DFGOT vol.19 (2003). Based on these the substance is classified as Category 2.
Serious eye damage/eye irritation	Moderate to severe irritation was noted and not recovered in 7 days, but completely recovered in 21 days in rabbit study (ECETOC TR 48 (1992)). Based on this the substance is classified as Category 2A. Vol.3, Risk Assessment, Ministry of the Environment (2004), SIDS (2004), EHC 65 (1987), ACGIH (2002), DFGOT vol.19 (2003) and recommendation of Japan Society for Occupational Health (1993) report that keratoconjunctivitis and such eye irritation were noted in human occupational exposure (vapour exposure). Classification is not possible due to lack of data.
Respiratory sensitization or skin sensitization	
Germ cell mutagenicity	The result of a somatic cell in vivo mutagenicity study, mammalian erythrocyte micronucleus test is negative (SIDS (2004), ACGIH (2002)). Based on this the substance is not classified.

Carcinogenicity	EPA classifies the substance as D. Based on this the substance is not classified.
Reproductive toxicity	According to Vol.3, Risk Assessment, Ministry of the Environment (2004), SIDS (2004), ACGIH (2002), DFGOT vol.19 (2003) and PATTY (4th, 1994), skeletal variation (cervical rib) in fetus only was noted in an inhalation exposure study in pregnant rat at the exposure concentration at which maternal toxicity was noted. The substance is not classified in accordance with the guidance.
Specific target organ toxicity (single exposure)	According to Vol.3, Risk Assessment, Ministry of the Environment (2004) and DFGOT vol.19 (2003), mild throat irritation was noted in human by inhalation exposure. Based on this respiratory tract irritation is suspected and the substance is classified as Category 3. Also, SIDS (2004), EHC 65 (1987), ACGIH (2002), DFGOT vol.19 (2003) and PATTY (4th, 1994) report that narcotic effect or suppression of central nervous system were noted in animal experiment, and vol.3, Risk Assessment, Ministry of the Environment (2004) and DFGOT vol.19 (2003) report that headache was noted in human by inhalation exposure. Based on these the substance is considered to cause narcotic effect and classified as Category 3.
Specific target organ toxicity (repeated exposure)	Vol.3, Risk Assessment, Ministry of the Environment (2004), EHC 65 (1987), ACGIH (2002), DFGOT vol.19 (2003), PATTY (4th, 1994) and recommendation of Japan Society for Occupational Health (1993) report that dizziness and headache were noted in human occupational exposure, and EHC 65 (1987), DFGOT vol.19 (2003), PATTY (4th, 1994) and recommendation of Japan Society for Occupational Health (1993) report that loss of auditory sense was noted. Based on these central nervous system and auditory organ are considered to be target organ and the substance is classified as Category 1.
Aspiration hazard	The product fulfills the condition of 'primary alcohol composed of carbon atom number with 3 or more and not more than 13, isobutyl alcohol or keton composed of carbon atom number with not more than 13'. Based on this the substance is classified as Category 2.
As N-Methylolacrylamide	
Acute toxicity: oral	LD50, rat = 563 mg/kg bw (NTP TR352 (1989)). Based on this the substance is classified as Category 4.
Acute toxicity: dermal	LD50, rabbit = about 16000 mg/kg bw (NTP TR352 (1989)). Based on this the substance is not classified.
Acute toxicity: inhalation (gas)	The substance is liquid under GHS criteria.
Acute toxicity: inhalation (vapour)	No data available.
Acute toxicity: inhalation (dust)	LC50, rat > 0.319 mg/L/6h (= 0.478 mg/L/4h) (IUCLID (2000)). This data alone is not sufficient to specify the classification and classification is not possible. The substance is regarded as dust because test concentration (0.319 mg/L) is greater than the saturated vapour pressure concentration (0.001 mg/L).
Acute toxicity: inhalation (mist)	LC50, rat > 0.319 mg/L/6h (= 0.478 mg/L/4h) (IUCLID (2000)). This data alone is not sufficient to specify the classification and classification is not possible. The substance is regarded as dust because test concentration (0.319 mg/L) is greater than the saturated vapour pressure concentration (0.001 mg/L).
Skin corrosion/irritation	“Mild to marked irritation” was noted in 24-hour application study on rabbit skin at 2000 – 16000 mg/kg (NTP TR352 (1989)). The dose level is quite high and the application period was longer than 4 hours. It is indicated that less irritation will be shown under the usual test conditions. Based on this the substance is not classified according to JIS classification criteria (corresponds to Category 3 of UN GHS classification).
Serious eye damage/eye irritation	“Mild to marked irritation” was noted in rabbit study (NTP TR352 (1989)). Based on this the substance is classified as Category 2.

Respiratory sensitization or skin sensitization	“Slightly irritating” result is reported in another application study at 3 mg in rabbit conjunctival sac (IUCLID (2000)). Respiratory sensitization: No data available. Skin sensitization: No data available.
Germ cell mutagenicity	The result is negative in bone marrow micronucleus test by oral or intraperitoneal administration in the mouse (somatic cell in vivo mutagenicity test) (NTP DB (Access on July. 2010)). Based on this the substance is not classified. As in vitro study negative result of Ames test and positive result of chromosome aberration study using CHO cells are reported (NTP DB (Access on July. 2010)).
Carcinogenicity	IARC classifies the substance as Group 3 (IARC 60 (1994)). Based on this the classification is not possible. No evidence of carcinogen was detected in rat, but increase of frequency of tumor in harderian gland, liver, lung and ovary was noted in mouse in 103-week oral dose administration study in the rat and mouse. The study concludes that the evidence of carcinogenicity is significant. (NTP TR352 (1989)). However IARC considers above knowledge from animal experiment is limited evidence (IARC 60 (1994)).
Reproductive toxicity	Remarkable systemic toxicity was not noted in F0 generation, decrease of litter size, prolonged interval between parturition, increase of postimplantation loss were noted in F0 and F1 generation in mouse two-generation reproductive oral administration toxicity study in accordance with protocol of reproductive assessment by continuous breeding (RACB). It is reported that exposure to the test substance caused moderate reproductive toxicity (NTP RACB 90017 abstract (1993)). It also reports that this substance causes dominant lethal effect and that all reproductive toxicity in F0 generation may be related to dominant lethal effect (NTP RACB 90017 abstract (1993)). Based on above, effect to reproduction is noted at the dose level which does not cause general toxicity to parental animals, and the substance is classified as Category 1B.
Specific target organ toxicity (single exposure)	It is reported that this substance caused peripheral nervous disorder to rat and mouse and the symptom is similar to that caused by acrylamide (IARC 60 (1994)). Ataxia, pain, loss of righting reflex, slight tonic-clonic convulsion before death were observed within LD50 range (350 - 510 mg/kg) after oral administration to the mouse (IUCLID (2000)). Based on this the substance is classified as Category 2 (nervous system). No adverse effect is reported at the dose level in the guidance range by inhalation and dermal administration (IUCLID (2000)).
Specific target organ toxicity (repeated exposure)	In 13-week repeated oral administration study in the rat and mouse, ataxia of hindlimb progressed to paralytic gait at 50 mg/kg/day and above in the rat, and in neuroethological examination, decrease of grip strength of fore- and hindlimb, startle response were noted at 12.5 mg/kg/day and above, increase of frequency of axonal degeneration of brainstem, spinal cord, peripheral nervous and degeneration of myelin sheath at 25 mg/kg/day and above (NTP TR352 (1989)). In mouse, decrease of grip strength of forelimb was noted at 25 mg/ kg/day and above, excessive startle response, decrease of motor coordination at 100 mg/kg/day (NTP TR352 (1989)). These effects were noted at the dose corresponding to Category 2 level of the guidance. Based on this the substance is classified as Category 2 (nervous system). In this rat 13-week repeated oral administration study in the rat, enlarged urinary bladder, inflammation, hemorrhage and oedema of mucous membrane of urinary bladder were noted and these are considered to be secondary effect (NTP TR352 (1989)). Therefore these findings are not taken for the basis of the classification.
Aspiration hazard	No data available.

As Acrylamide	
Acute toxicity	Oral, LD50, mouse = 107 mg/kg Oral, LD50, rat = 124 mg/kg Oral, LD50, rabbit = 150 mg/kg
Acute toxicity: oral	LD50 = 124 mg/kg (EHC 49 (1985)), 175 mg/kg (EU-RAR No.24 (2002)), 203 mg/kg (EU-RAR No.24 (2002)) in oral administration study in the rat. Calculation was used and the result is LD50 = 129 mg/kg. Based on this the substance was classified as Category 3.
Acute toxicity: dermal	LD50 = 400 mg/kg (EHC 49 (1985)), 1148 mg/kg (EU-RAR No.24 (2002)) in rat dermal administration study. Based on this the substance is classified as Category 3 according to the lower LD50 = 400 mg/kg.
Acute toxicity: inhalation (gas)	The substance is solid under GHS criteria. Gas inhalation is not expected and the classification is not applicable.
Acute toxicity: inhalation (vapour)	No data available.
Acute toxicity: inhalation (dust)	No data available.
Acute toxicity: inhalation (mist)	No data available.
Skin corrosion/irritation	Based on data of skin irritation study in the rabbit (EU-RAR No.24 (2002)) and effect to human health (EU-RAR No.24 (2002), ACGIH (7th, 2005)), it is considered that the substance is "mild irritant". Therefore the substance is classified as Category 3.
Serious eye damage/eye irritation	Eye irritation study in the rabbit (EU-RAR No.24 (2002)) reports "irritant effect to eye was noted and disappeared in 21-day observation period". Based on this the substance is classified as Category 2A.
Respiratory sensitization or skin sensitization	Respiratory sensitization: No data available. Skin sensitization: Based on positive result of skin sensitization study using the guinea-pig (EU-RAR No.24 (2002)), it is considered the substance is skin sensitizer and the substance is classified as Category 1.
Germ cell mutagenicity	Positive in germ cell in vivo heritable mutagenicity studies (dominant lethal test and specific locust test) according to CERI/NITE Hazard Assessment Report No.35 (2004). Based on this the substance is classified as Category 1B.
Carcinogenicity	IARC (1994) classifies the substance as 2A, U.S. NTP (2005) classified as R. Based on these the substance is classified as Category 1B.
Reproductive toxicity	Decrease of litter size, lowered fertility of male animals, increase of post- and pre-implantation loss were noted at the dose level at which general toxicity was not noted according to CERI/NITE Hazard Assessment Report No.35 (2004).
Specific target organ toxicity (single exposure)	For human, "hallucination, hypotension, epileptic seizure and peripheral neuropathy were observed, and peripheral neuropathy was observed also at 2 month observation" (EU-RAR No.24 (2002)). For experimental animals, "tremor, hyperreflexia, repetitive tonic-clonic convulsion, lack of coordination of hind limb, lack of coordination of posture and motor, functional disorder of hind limb muscle, and vacuolation of spermatid nucleus" (EU-RAR No.24 (2002)) were observed. Based on these it is considered that nervous system (peripheral and central nervous system) mainly and testis are target organs. Recovery of neurological disorder is considered to be poor. Effects on experimental animals are observed in the guidance range of Category 1 level. Based on above, the substance is classified as Category 1 (nervous system, testis).
Specific target organ toxicity (repeated exposure)	For human, "gait disturbance, memory impairment, hallucination, speech disturbance, numbness of limbs, abnormal sweating of hands and feet, and dysgeusia" (CERI/NITE Hazard Assessment Report No.35 (2004)), and "muscle weakness, lack of coordination of upper limb, tremor of hands, sensory disturbance (vibratory sense, sense of pain, sense of touch, sense of position), decreased tendon and sole

reflex, nystagmus, diffuse fibrosis of nerve fiber, and swelling of axis with disappearance and demyelination” (EU-RAR No.24 (2002)) are reported. For experimental animals, “decrease of sense of balance, lowered vision, weakness of hind limb, tremor of upper limb, axonal degeneration of peripheral nerve, axonal degeneration of optic tract fiber, damage of spinal cord, testicular atrophy, decrease of spermatid and spermatocyte, and decrease of sperm count” (CERI/NITE Hazard Assessment Report No.35 (2004)) are reported. It is considered that nervous system and testis are target organs. Effects on experimental animals are observed in the guidance range of Category 1 level. Based on above, the substance is classified as Category 1 (nervous system, testis).

Aspiration hazard No data available.

12.ECOLOGICAL INFORMATION

As N-n-Butoxymethyl acrylamide

Aquatic hazard (acute) No data available.

Aquatic hazard (long-term) No data available.

As n-Butanol

Aquatic hazard (acute) Fish (killifish), 96hr LC50 > 100 mg/kg (Ecotoxicity study, Ministry of the Environment, 1996) Based on this the substance is not classified.

Aquatic hazard (long-term) Not poorly water soluble (water solubility = 63200 mg/L (PHYSPROP Database, 2005)) and low level of acute toxicity. Based on this the substance is not classified.

Ecotoxicity LD50, golden fish, 24h TLm >= 1900 mg/L

Persistence/degradability [Degradability] The residual amount was 0 at day 2 in a study in accordance with BOD test method in JIS. Percentage degradation by oxygen consumption demand for 5 days was 60%.

As N-Methylolacrylamide

Aquatic hazard (acute) Fish (rainbow trout), 96hr LC50 = 356 mg/L (IUCLID, 2000). Based on this the substance is not classified.

Aquatic hazard (long-term) Not classified for acute toxicity and not poorly water soluble (water solubility = 653 g/L (PHYSPROP Database, 2011)). Based on this the substance is not classified.

As Acrylamide

Aquatic hazard (acute) Algae (green alga), 72hr EC50 = 16.9 mg/L (EU-RAR, 2002). Based on this the substance is classified as Category 3.

Aquatic hazard (long-term) Rapidly biodegradable (percentage biodegradability by BOD: 70% (Existing Chemical Substance Safety Assessment)) and also estimated to be low level of bioaccumulation (log Kow = -0.67 (PHYSPROP Database, 2005)). Based on this the substance is not classified.

13.DISPOSAL CONSIDERATIONS

Residual waste Dispose of the substance by contracting an agency for industrial waste disposal licensed by the prefectural governor or a local public organization for industrial waste disposal, if available.

Polluted container and packing Containers must be recycled after cleaning or disposed of in accordance with the related laws/regulations or municipal codes. Remove contents completely before the disposal of empty containers.

14.TRANSPORT INFORMATION

International regulations

Marine regulations	Follow the rules and regulations under IMO.
UN No.	2810
Proper shipping name	TOXIC LIQUID,ORGANIC,N.O.S. (N-n-BUTOXYMETHYLACRYLAMIDE,STABILIZED)

Class	6.1
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Packing group	III
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Marine pollutant	Not applicable.
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Transport in bulk according to	Not applicable.
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MARPOL 73/78, Annex II, and the IBC code

Air regulations	Follow the rules and regulations under ICAO/IATA.
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UN No.	2810
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Proper shipping name	TOXIC LIQUID,ORGANIC,N.O.S. (N-n-BUTOXYMETHYLACRYLAMIDE,STABILIZED)
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Class	6.1
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Packing group	III
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Special safety precautions during transport	Enforce the preventive measures against the fall of the hazardous substance or the fall, inversion or damage of the containers of the hazardous substance.
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Avoid applying vigorous friction or shaking to the hazardous substance or containers during transportation.

Protect from direct sunlight during transportation. Load the container by enforcing preventive measures against damage, corrosion, or leakage from the container, as well as against load collapse.

15.REGULATORY INFORMATION

Follow all regulations in your country.

16.OTHER INFORMATION

Contact	Send inquiry about the contents of this SDS to: Shinryo Corporation Marketing Division: +81-93-643-2778
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References	Database of Japan Chemical Database Ltd. SDS of the raw materials used in the product. "Prehled Prumyslove Toxikologie; Organicke Latky," Marhold, J., Prague, Czechoslovakia, Avicenum, 1986Vol. -, Pg. 706, 1986.
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Others	The descriptions herein are based on the currently available references and sources. Data may be revised on obtaining the new information. All precautions described herein are for normal handling. Please use the information herein on the user's own responsibility for the decision of the required safe use. This SDS provides information only and no guarantee is given to the contents herein.
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